Explainable Shipment Route Recommendation under Cost, Time and Risk

Technical report | Aiqon AI Internship take-home | Janagam Vignesh | code: see GitHub repository (README has run instructions)

1. What "optimal" means here

A route is optimal for a shipment if it is the best match for that shipment's priority **among routes that satisfy its non-negotiable requirements**. I separate the two kinds of requirement on purpose: **hard constraints** are must-haves a planner would never trade away (a minimum probability of meeting the deadline, a cap on route risk); **soft objectives** (freight cost, transit time, lateness risk, route risk) are traded off with weights that depend on whether the shipment is urgent, standard or cost-sensitive. This mirrors how planners work, keeps the logic auditable, and avoids the failure mode of a pure weighted sum, where a large enough saving can buy a route that will almost certainly miss the deadline.

2. Synthetic data and how trade-offs were constructed

40 shipments over 10 lanes (Asia-Europe, trans-Pacific, India-Middle East/Africa), 142 routes (3-5 per shipment, seeded and reproducible). Every shipment always has the cheap congested option, and the other archetypes appear randomly, so each shipment presents a genuine dilemma. Cargo value (log-normal, $20k-$600k), weight, priority (25% urgent / 45% standard / 30% cost-sensitive) and deadline (a priority-dependent multiple of the direct route's transit time) vary per shipment. High-value cargo gets a stricter risk cap (0.5 vs 0.7).

Route archetype	#	Avg cost	Avg days	Delay prob.	Geo risk	Trade-off it encodes
Ocean - direct	40	$2,002	19.1	0.41	0.17	Baseline: cheap, congested hubs, slow, often late
Ocean - reliable hubs	35	$2,155	21.1	0.14	0.14	1.15x direct cost, safe hubs, much lower delay probability
Ocean - chokepoint shortcut	22	$1,681	15.8	0.21	0.78	0.92x cost and quicker, but via Red Sea/Hormuz/Black Sea: high geopolitical risk
Sea-air combo	23	$4,836	9.5	0.14	0.13	2.3x cost, much faster, fairly reliable
Air express	22	$10,380	2.8	0.06	0.10	4.8x cost, fastest and most reliable

Averages pool different lanes; multipliers in the last column are the nominal design values.

True vs observed. I first draw true parameters, then give the recommender only noisy *estimates* (delay probability +/-0.06, transit time +/-6%, risk indicators +/-0.08-0.10; freight quotes are exact). The true values are used only by the evaluation harness, so the system is judged on what would actually happen, not on its own inputs.

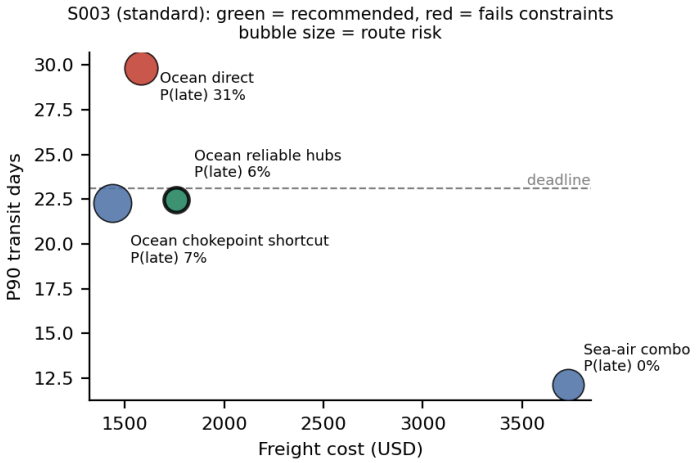
3. Decision framework

- **Uncertainty-aware metrics.** Each route is simulated 2,000 times: a delay occurs with the route's delay probability (gamma-distributed length), a geopolitical disruption occurs with probability 0.10 x geo-risk (+7 days, 15% cargo loss), and the delay/risk *estimates themselves* are jittered. Output: P(on time), P90 transit days, expected total cost (freight + holding + delay + SLA penalty + disruption loss).
- **Hard constraints.** P(on time) >= 85% / 70% / 60% (urgent / standard / cost-sensitive) and composite risk (0.4 x congestion + 0.6 x geopolitical) below the cargo's cap. If *nothing* qualifies, the system says so, drops the reliability constraint and ranks risk-compliant routes by expected total cost, because probability alone cannot tell 0.4 days late from 10 days late.
- **Pareto filter.** Routes that are no better than another eligible route on cost, P90 time, P(late) and risk are removed.
- **Weighted score.** Score = sum of weight x penalty, lower is better. Penalties live on absolute, decision-relevant scales: cost = premium over the cheapest route (+100% scores 1), time = P90 / deadline, lateness = P(late), risk = composite risk. Weights (cost/time/late/risk): urgent: 0.15/0.30/0.35/0.20; standard: 0.35/0.20/0.25/0.20; cost_sensitive: 0.55/0.10/0.15/0.20. They are preference judgements and are exposed in the CLI and the demo.
- **Why absolute scales?** My first version used min-max normalisation. Evaluation showed that one expensive air option stretched the cost scale, making all sensible ocean options look equally cheap, so the system overpaid for speed (regret 20.5% vs 10.0% now; avg freight $3,972 vs $2,758). It is kept as an ablation.
- **Why not a learned model?** There are no outcome labels for "best route", and an opaque model would be hard for an operator to challenge. A transparent framework plus simulation is more useful here.

4. Explanations an operator can use (shipment S003, standard, deadline 23 days)

```
RECOMMENDATION: R009 - Ocean - reliable hubs (Shanghai > Salalah > Los Angeles)
$1,760 freight | ~20.5 days (P90 22.4) | 94% chance on time | risk 0.15 | expected total cost $2,783
Shipment is 'standard': must arrive within 23 days with >= 70% confidence and route risk <= 0.70.
WHY THIS ROUTE
- vs the cheapest eligible route R010: this route costs $322 (22%) more; has lower route risk (0.54 -> 0.15).
- vs the fastest eligible route R011: this route costs $1,971 (53%) less; raises the chance of missing the deadline from 0% to
6%; is 10.5 days slower (P90); has lower route risk (0.33 -> 0.15).
- Closest alternative is R010; the deciding factor is lower route risk.
ROUTES NOT RECOMMENDED
- R008 Ocean - direct: only 66% chance of meeting the 23-day deadline (needs at least 70%)
CONFIDENCE: 53% of 300 stress tests (weights and estimates perturbed) pick this route; main alternative R010 (47%). Decision
stability: LOW - a judgement call.
```

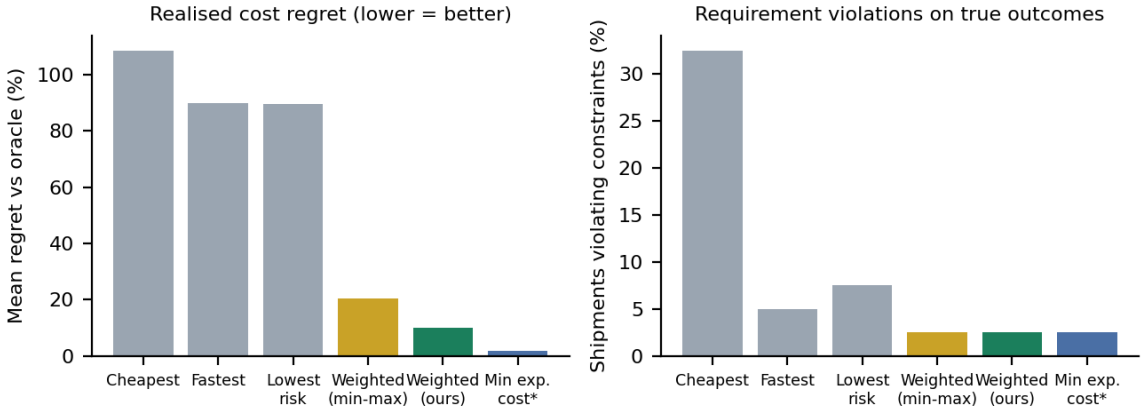
Every output names the winner, compares it with the cheapest and fastest eligible alternatives in plain trade-off language ("costs $X more but cuts P(late) from a to b"), states the deciding factor, lists why each other route was excluded, and ends with a confidence line from the stress test in section 6.



5. Evaluation

Ground truth is a 4,000-run simulation of each route using its *hidden true* parameters. The **oracle** is the route with the lowest true expected total cost (unattainable in practice). **Regret** = how much more the chosen route costs than the oracle, averaged over shipments. **Violations** = chosen route's true P(on time) or risk breaks the shipment's requirement.

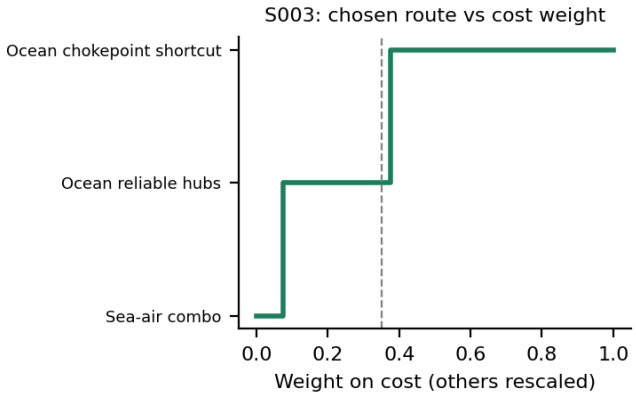
Policy (sees only observed data)	Mean regret	Matches oracle	True P(on time)	Req. violations	Avg freight	Avg days
Always cheapest	108.5%	12%	74%	32.5%	$1,901	20.7
Always fastest	89.7%	22%	97%	5.0%	$7,199	6.8
Always lowest risk	89.4%	48%	92%	7.5%	$5,494	11.9
Weighted, min-max scaling (ablation)	20.5%	57%	96%	2.5%	$3,972	15.4
Constraints + Pareto + weighted (ours)	10.0%	65%	94%	2.5%	$2,758	18.2
Min expected total cost*	1.7%	85%	94%	2.5%	$2,589	18.7

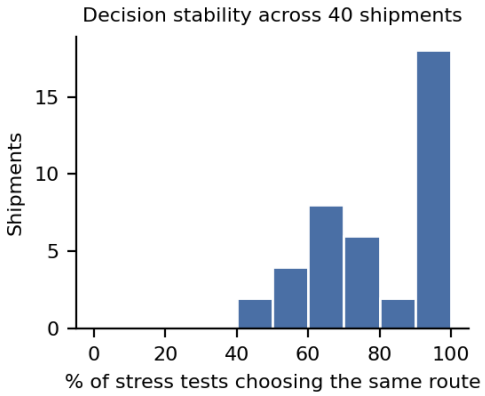


- Single-metric rules fail in predictable ways: *cheapest* breaks a requirement for 32% of shipments (on-time only 74%); *fastest* and *lowest risk* pay $7,199 and $5,494 on average, about 2.6x and 2.0x our $2,758, giving ~90% regret.
- The framework cuts regret to 10.0% with 94% true on-time probability. Its one violation is S007, an urgent shipment with a 5-day deadline that no available route can meet; the system flags it rather than hiding it.
- ***Honest caveat:** minimising expected cost scores best (1.7% regret) because my ground truth uses the same cost model, so this comparison is partly circular and I treat it as a reference, not a competitor. In practice dollar penalties for lateness are rarely known precisely, which is why the weighted approach (needing only stated preferences) is the default and expected-cost ranking is offered as an option.

6. Uncertainty and sensitivity

- **Stress test per shipment (300 runs):** weights are redrawn around the profile (Dirichlet) and cost, time, P(late) and risk estimates are perturbed. Mean stability is 80%; 47.5% of shipments are at least 85% stable and 15.0% fall below 60% and are labelled "a judgement call" in the output, which is the honest message when two routes are close.
- **Weight sweep:** for S003 the choice flips from the reliable-hub route to the chokepoint shortcut when the cost weight rises just above its default (figure below), exactly the kind of knife-edge an operator should be told about.
- **Estimate errors:** scaling all delay probabilities by -25% / +25% changes 0.0% / 2.5% of recommendations; scaling geopolitical risk by -25% / +25% changes 15.0% / 7.5%. Decisions are robust to delay-estimate error but more sensitive to geopolitical risk, because chokepoint routes sit near the risk caps.





7. Assumptions and limitations

- **Synthetic data.** Costs, speeds, delay rates and the economic parameters (delay cost per day, SLA penalties, 10% disruption rate) are assumed, not calibrated; absolute numbers are illustrative and only the relative behaviour is meaningful. With 40 shipments, regret estimates are noisy.
- **Independence.** Legs and routes fail independently; real disruptions are correlated (a Red Sea closure hits every route using it). A scenario-based or CVaR objective would capture tail risk better than P90.
- **Subjective inputs.** Weights, caps and the +100% cost scale are judgements. They are exposed and stress-tested, but should be elicited from planners (e.g. pairwise comparisons) in a real deployment.
- **Scope.** No capacity or booking availability, no shipment consolidation, no mid-journey re-routing; recommendations are a one-time snapshot.
- **Next steps.** Calibrate delay/risk from port-performance and schedule-reliability data, add correlated disruption scenarios, learn weights from planner overrides, and log decisions to monitor drift.